

TEMPERATURE MONITORING On Smartphone

SAIKAT PATRA AND SHIBENDU MAHATA

This article describes a temperature monitoring system on an Android smartphone. Temperature is sensed by a temperature sensor installed in the circuit, and is sent over Bluetooth to the smartphone. An Android application is used in the smartphone to display temperature data. The system provides easy monitoring of process data on a smartphone.

You can connect the smartphone wirelessly with the system and monitor the process variable, when desired. The system can easily be customised to monitor other process variables also. Authors' prototype of the transmitter side is shown in Fig. 1. The receiver side consists of the smartphone. A screenshot of the Android app is shown in Fig. 2. Block diagram of the project is shown in Fig. 3.

Circuit and working

The circuit diagram of the temperature monitoring system on smartphone is shown in Fig. 4. Components used in this project are given below.

Android smartphone. Install the temperature monitoring app on Android smartphone. The app (temperaturemonitoring.apk) used here has been built using MIT App Inventor. When the app is running on the smartphone, data sent by HC-05 Bluetooth module is received and displayed on the app.

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and 14 digital I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI and I2C. The MCU can operate at a clock frequency of 16MHz.

On the transmitter side, interface LM35 temperature sensor with

Arduino Uno.

HC-05 Bluetooth module. HC-05 module is an easy-to-use Bluetooth Serial Port Protocol (SPP) module, designed for transparent wireless serial connection setup. Serial port Bluetooth module is a fully-qualified Bluetooth V2.0 + EDR (enhanced data rate) 3Mbps modulation with complete 2.4GHz radio transceiver and baseband. It uses CSR Bluecore 04-external single-chip Bluetooth system with CMOS technology and adaptive frequency hopping feature. It can auto-connect to the last device on power by default. The default auto-pairing password is 1234.

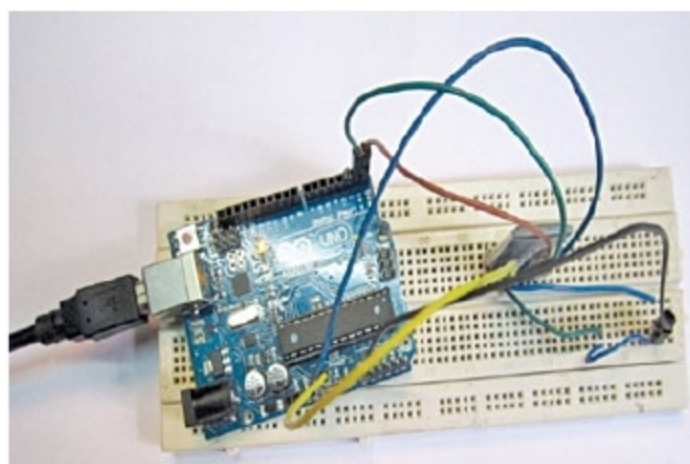


Fig. 1: Authors' prototype on the breadboard



Fig. 2: Screenshot of Android application

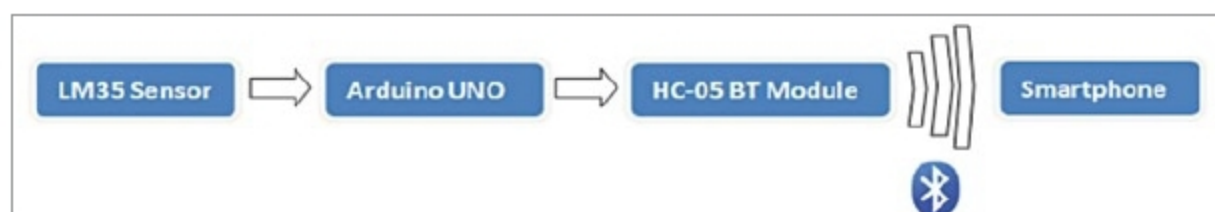


Fig. 3: Block diagram of the project

LM35 temperature sensor. LM35 series are precision integrated circuit (IC) temperature sensors, whose output voltages are linearly proportional to Celsius (Centigrade) temperature. LM35, thus, has an advantage over linear temperature sensors calibrated in ° Kelvin. You are not required to subtract a large constant voltage from its output to obtain convenient Centigrade scaling.

LM35 does not require any external calibration or trimming to provide typical accuracies of $\pm 1/4^\circ\text{C}$ at room temperature and $\pm 3/4^\circ\text{C}$ over a full -55 to $+150^\circ\text{C}$ temperature range. Scale factor is $10\text{mV}/^\circ\text{C}$. Vout pin of LM35 temperature sensor is connected to analogue input pin A0 of Arduino Uno on the transmitter side.

On the transmitter side, LM35 temperature sensor is used to sense temperature. Arduino senses this voltage and the program (Temp.ino) running in the MCU calculates the equivalent temperature. Temperature value is sent to HC-05 Bluetooth module via serial communication.

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno
- IC1 - LM35 temperature sensor

Miscellaneous:

- BATT.1 - 9V rechargeable battery or power adaptor
- HC-05 Bluetooth module
- Android phone
- Male-to-female jumper wires